

CS101 - Introduction to Computer Science

Midterm Exam

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Instructions: Answer all questions. Show your work where applicable.

Q1. (10 pts) Explain the difference between a stack and a queue. Give a real-world example of each.

A stack is LIFO (Last In First Out) - like a pile of plates.

You add and remove from the top only.

A queue is FIFO (First In First Out) - like a line at a shop.

People join at the back and leave from the front.

Q2. (10 pts) What is the time complexity of binary search? Explain why.

Binary search is $O(\log n)$.

Each step we cut the search space in half.

So for $n=1000$ we need about 10 steps only.

This is much faster than linear search $O(n)$.

Q3. (10 pts) Define recursion and write a recursive function to calculate factorial of n .

Recursion is when a function calls itself to solve a smaller problem.

```
def factorial(n):
```

```
    if n == 0: return 1
```

```
    return n * factorial(n-1)
```
